

Modulating Huntingtin Exon-1 Protein Structural Motifs with Flanking Domains

Francisco X. Vázquez^{1,2}, Hongsuk Kang¹, Leili Zhang¹, Leticia Toledo-Sherman³, Johannes Voigt³, Binqun Luan¹, and Ruhong Zhou^{1,4,*}

1. Computational Biology Center, IBM Thomas J. Watson Research Center, Yorktown Heights, NY 10598, USA

2. Department of Chemistry, St. John's University, Queens, NY 11439, USA

3. CHDI Management/CHDI Foundation, Los Angeles, CA 90045, USA

4. Department of Chemistry, Columbia University, New York, NY 10027, USA

Abstract

There is an increasing interest in the role of the domains that flank the extended polyglutamine (polyQ) region in huntingtin (Htt) exon-1 protein; especially their role in the protein's aggregation and fibril formation. The polyQ is flanked by a 17-residue N-terminal amphipathic domain (N17) and a 38-residue mostly polyproline domain (C38). An important question regarding the aggregation of Htt is how these flanking domains modulate the structural motifs associated with the formation of fibrils. Previous experimental studies have indicated that the C38 domain can destabilize fibril formation, while the N17 domain enhances fibril formation and destabilizes non-fibril forming aggregates. To explore the molecular mechanisms behind these observations, we used temperature replica exchange molecular dynamics (REMD) simulations to investigate Htt variants with either N17, C38, or both deleted, and analyzed how these flanking domains affect the structural and polymeric properties of Htt. Structural analyses of Htt with the C38 domain deleted indicate an increase in β -sheet formation, suggesting that the C38 domain plays a protective role. Furthermore, the removal of the N17 domain leads to a reduction in β -sheet content, which correlates well with N17's general role as a stabilizer of fibril formation. More specifically, we found that interactions between the N17 and the polyQ domain lead to the formation of well-formed β -hairpins in both domains. This suggests that the N17 is able to stabilize β -sheet motifs on the polyQ domain that may play an important role in forming Htt fibrils. We also found that the removal of the C38 domain reduces the radius of gyration in the polyQ domain and leads to less protein flexibility. All together, these results imply, consistent with the experimental findings, that the N17 domain may act as a monomeric β -sheet template to help destabilize non-fibril forming aggregates while the C38 acts to destabilize β -sheet formation.

Introduction

Huntington's disease (HD) is caused by a mutational CAG expansion in the exon-1 region of the huntingtin gene¹. This mutation results in an expanded polyglutamine (polyQ) region in the N-terminal region of the huntingtin (Htt) protein. The wild type, nonpathogenic polyQ region has 22 glutamine (Q) residue repeats and the pathogenic-threshold length for HD is 36-40¹⁻². It has been shown that the age of symptom onset for HD is inversely related to the length of the polyQ region²⁻⁴. Although the relationship between the CAG repeat length (Q-length) and disease progression is well established, the underlying molecular mechanism is still poorly understood.

The exon-1 region of the Htt protein contains two other distinct flanking domains: an amphipathic N-terminal domain made of 17 residues (N17) and a 38 residue C-terminal domain (C38) that contains two polyproline (polyP) chains connected by a hinge region. The amphipathic nature of the N17 domain causes it to be helical in lipid bilayers and unstructured in aqueous solution, which has been proven experimentally with solution NMR⁵ and theoretically with molecular dynamics (MD) simulations⁶⁻⁸. The N17 domain has also been found to adopt a helical structure in micelles and hydrophobic nanomaterials^{7, 9-12}. On the other hand, the polyP chains of the C38 domain have been shown to form rigid poly-L-proline type II (PPII) helices both experimentally¹³⁻¹⁴ and computationally¹⁵.

Recent experimental work has demonstrated that the mutant Htt forms oligomeric fibrils through interactions between the **β-sheet structures**¹⁶⁻²⁰. Even though the toxicity of the fibrils is controversial²¹, they are known to be the hallmark of patients with HD²². Additionally, mutant Htt variants can also form toxic non-fibril forming aggregates. The flanking domains have been found to regulate the formation of fibrils and non-fibril forming aggregates. Without them, the polyQ region is prone to aggregation even with shorter, non-pathogenic Q-lengths²²⁻²⁸. The C38 domain has been shown to destabilize fibril formation²⁸⁻²⁹, serving a protective role by suppressing the formation of toxic oligomers. The N17 domain, however, has been shown to promote fibrilization³⁰⁻³¹, but also destabilize non-fibril forming aggregates when the C38 domain is removed^{28, 32-33}. Although these previous studies have helped to elucidate some of the mechanisms behind these observations, it is still unclear how the flanking domains influence the behavior of the monomer, and how that may lead to the modulation of Htt protein aggregation.

Computational work by Pappu and coworkers has suggested that removing the C38 domain allows the N17 domain to become entangled with the polyQ region³²⁻³³. This may explain why the removal of the C38 domain, with only N17 present, leads to a destabilization of non-fibril forming aggregates. Using a solvent free force field designed to simulate intrinsically disordered proteins (IDPs)³⁴, they were able to reveal entangled structures within the domains and simulate protein oligomers³²⁻³³. Through a combination of computational and spectroscopic work, they found an increase in the helical content of the N17 region; however, they did not find an enhancement in β -content involved in dimerization. This is in contrast to previous experimental work by van der Wel and coworkers that has indicated that β -sheet content is important to fibril formation¹⁷. Pappu and coworkers proposed that β -sheet formation occurs after oligomerization within the oligomers³². Recent computational work by our group, however, has demonstrated that increased Q-length leads to increases in β -sheet content in the monomer¹⁵.

In this work, we sought to reveal the detailed interactions among various domains of the Htt exon-1 protein that hopefully lead to a clearer molecular picture for the observed roles of the flanking domains. To this end, we simulated three Htt exon-1 variants, with either N17, or C38, or both, deleted: Q₄₀C38, N17Q₄₀K₂, and Q₄₀K₂, respectively. By comparing these domain deletion variants with the full Htt exon-1 systems, N17Q₂₂C38, N17Q₄₀C38 and N17Q₅₆C38 (corresponding to the non-pathogenic, the pathogenic threshold, and a fully pathogenic Q-length, respectively) that were previously simulated¹⁵, we are able to show that the removal of the flanking domains changes the propensity of the protein to form secondary structure motifs, as well as changes in the compactness of the protein. We have found that the removal of the C38 domain leads to increased β -sheet content. We also found that without the C38 domain, the N17 increases the number of contacts it forms with the polyQ domain, which can lead to β -sheet structures being formed between the N17 and polyQ regions that further stabilize the β -sheets within the polyQ region. In analyzing our previous full protein simulations, we found that interactions between the N17 and C38 tend to stabilize structures that are less compact and have less β -sheet content. These results suggest that the protective nature of the C38 domain might be due to its ability to form contacts with both the N17 and polyQ region in a way that can keep them from interacting with each other.

Results and Discussion

Removal of the C38 domain leads to increased β -sheet content

The full Htt protein has 3144 amino-acid residues and many different domains, including structured HEAT motifs and disordered regions. Disease activity of the protein is linked to the N-terminal exon-1 region, which consists of the N17, the polyQ, and the C38 domains (**Fig. 1a**). The exon-1 protein is inherently disordered, as shown in **Fig. 1b**, even though both the compactness of the protein and the propensity to adopt secondary structure motifs -especially β -sheet content¹⁵- increases with Q-length . This increase in β -sheet content may lead to further aggregation and subsequent fibrillization, which could be the source of the toxicity seen at higher Q-lengths ^{26, 35-36}.

Our simulations revealed that the removal of the C38 domain leads to an increase in the β -sheet content of the protein. **Fig. 1c** shows relatively high-probability structures derived from clustering analyses applied to the polyQ region of the N17Q₄₀K₂, Q₄₀K₂ and Q₄₀C38 protein variants. The clusters were found using the GROMACS 5.0.5 cluster analysis tool³⁷ based on the algorithm developed by Daura *et al* ³⁸, applying a 3.5 Å RMSD cut-off. The N17Q₄₀K₂ cluster center shows that when the C38 domain is removed from an N17Q₄₀C38 exon-1 protein, the protein is able to adopt β -hairpin like structures in the polyQ domain. These structures are also reinforced by the β -bridge structures (shown as beads in the figure) on the N17 domain that fluctuate between the bridge and sheet motifs. In these structures, the N17 domain appears to help stabilize the well-formed β -sheet motifs in the polyQ domain. In the case of the Q₄₀K₂ variant, the elimination of the C38 domain again leads to an increase in the β -sheet content. The β -sheets formed by the Q₄₀K₂ variant are larger than the N17Q₄₀K₂ protein but not as well-formed (**Fig. 1c**). When the N17 domain is removed but the C38 domain is retained, as in the Q₄₀C38 case, the structure representing the cluster center shows very little β -content in the polyQ domain. The polyP region generally adopts a rigid PPII helical structure. We see from our simulations that this motif interacts with the polyQ domain, which appears to destabilize the β -sheet content in the protein. It appears that, through its interactions with the polyQ region, the C38 domain may play a protective role in the polyQ aggregation.

More quantitative measures also reveal that the removal of the C38 domain leads to a significant decrease in the β -sheet content. **Fig. 2** depicts plots of the average number of residues adopting different secondary structure motifs determined by STRIDE³⁹ for each full length and domain

deletion variant of the Htt exon-1 protein. As shown in **Fig. 2a**, the entire N17Q₄₀K₂ protein samples significantly more β -sheet content than even the N17Q₄₀C38, N17Q₅₆C38, or Q₄₀K₂ variants. Due to an increase in the β -sheet content from both the polyQ and N17 regions, the N17Q₄₀K₂ variant has an average of four residues adopting the β -sheet conformation, which is even ~ 1.5 residues more than the N17Q₅₆C38 case (in general, the full exon-1 protein shows increases in β -sheet content with increases in Q-length¹⁵). **Fig. 2b** shows that the majority of the β -sheet content in the N17Q₄₀K₂ protein comes from the polyQ domain, even though the N17 domain also shows meaningful contributions (**Fig. 2c**). This elevation of the β -sheet content is at least partially due to the formation of structures such as those seen in **Fig. 1c**, where β -type structures are formed between the N17 and polyQ regions that can result in increased stabilization of β -type motifs.

Conversely, the Q₄₀C38 variant was found to have much less β -sheet content than the Q₄₀K₂ protein, suggesting that interactions between the polyQ and C38 domains can destabilize the β -sheet formation. This destabilization of β -sheet content may provide a possible explanation for the experimental observations from Pappu and coworkers²⁸ that Q₄₀C38 can lead to a kinetic destabilization of fibril formation. Lastly, **Fig. 2d** shows a slight decrease in the β -sheet content in the C38 domain; however, the total amount of β -sheet content in the C38 domain is very small, likely due to the fact that secondary structures are rarely formed within the hinge region between the two polyP chains.

Although binary secondary structure assignment algorithms, such as STRIDE, provide a good measure of the number of residues that adopt well-defined secondary structure motifs, it can be difficult to capture the transient or imperfect secondary structure motifs that often exist in IDPs. Because of this, we used the secondary structure scoring functions developed by Pietrucci and Laio⁴⁰ that are based on the root-mean squared differences between a segment of residues in the protein and an idealized β -sheet or α -helix. For these particular scoring functions, a score of one means that six residues in the protein (in sequence for the α -helix score or two groups of three in sequence for the β -sheet score) perfectly match an idealized structure. We used the β -sheet and α -helix secondary structure scores as collective variables to calculate a two-dimensional map of potentials of mean force (PMF) for the polyQ domain of the N17Q₂₂C38, N17Q₄₀C38,

N17Q₅₆C38, N17Q₄₀K₂, Q₄₀K₂, and Q₄₀C38 variants (**Fig. 3**). In each of the PMF plots, a dotted line shows where the β -sheet score is equal to 1.0.

The top panels of **Fig. 3** show the secondary structure PMF for the polyQ region in N17Q₂₂C38, N17Q₄₀C38, and N17Q₅₆C38 proteins. As expected from our previous simulation studies¹⁵, the β -content increases with the Q-length faster than the helical content. As the Q-length increases, more structures are sampled with a β -sheet score greater than one, suggesting the formation of larger amounts of β -content in the protein. The non-pathogenic “wild-type” N17Q₂₂C38, specifically, mostly samples structures with a β -sheet score close to zero. The N17Q₅₆C38 variant, however, shows an energy minimum well where structures with β -sheet scores of ~ 1.5 are sampled, in addition to other wells with a significant amount of β -content.

The bottom panels of **Fig. 3** show the secondary structure PMFs of the polyQ domain for the domain deletion variants, N17Q₄₀K₂, Q₄₀K₂, and Q₄₀C38. For the N17Q₄₀K₂ protein, the two most prominent energy minimum wells lie where the β -sheet score is ~ 1.0 . This is due to the well-formed, β -hairpin like structures formed in the polyQ domain and stabilized by the β -bridges on the N17 region, as seen in **Fig. 1c**. Even though N17Q₄₀K₂ showed greater β -sheet content in the polyQ when measured with STRIDE (**Fig. 2b**), the polyQ PMF for Q₄₀K₂ shows that it is able to sample more overall β -content than N17Q₄₀K₂ does. This is likely due to the more transient and less-stable nature of the β -structures sampled by Q₄₀K₂. Additionally, Q₄₀K₂ may be sampling more structures that form imperfect β -sheet-like structures. These imperfect β -sheet-like structures may not be captured by binary methods like STRIDE, but do contribute to the overall β -sheet score in the Pietrucci and Laio method⁴⁰.

The Q₄₀C38 PMF shows that the removal of the N17 region leads to increased sampling of structures with reduced β -sheet content. Comparing the Q₄₀C38 and Q₄₀K₂ PMFs, the removal of the C38 region shifts the location of the PMF wells away from where the β -sheet score is ~ 1.0 . The C38 domain appears to significantly destabilize the β -sheet content, suggesting that it may play a protective role against aggregation and fibrillization of Htt exon-1 proteins. This is consistent with the previous experimental findings that Q₄₀C38 leads to reduced fibril formation²⁸⁻²⁹.

Removal of C38 domain increases polyQ compactness

In our previous work¹⁵, we were able to show that increases in the Q-length lead to more compact polyQ structures. This increase in the compactness was associated with an increase in the amount of β -sheet content in the polyQ domain¹⁵. In the current work, we found that the removal of the C38 domain (N17Q₄₀K₂ and Q₄₀K₂) also leads to an increase in the compactness of the polyQ domain. As shown in **Fig. 4a**, the calculated radius of gyration (R_g) of the polyQ domain (only the polyQ region is included in the calculation) for both the N17Q₄₀K₂ and Q₄₀K₂ proteins show similar distributions with tight variances. The Q₄₀C38 and N17Q₄₀Q38 proteins show broader distributions with more conformations arising from less compact structures. The N17Q₄₀C38 protein also shows a small peak at a larger R_g value that is not sampled by any of the deletion variants, noted on the plot by the dashed box. The right side of **Fig. 4a** shows the cluster center of the structures sampled in the dashed box. In the sampled area, the N17 and C38 domains interact with each other to sandwich the polyQ region and stabilize the elongated structures, thus helping to modulate the compactness of the polyQ domain.

To compare the compactness of the domain deletion variants to the full length Htt exon-1 proteins (**Fig. 4b**), we also plotted the distributions of the normalized radius of gyration, R_g^* of the polyQ region for the N17Q₂₂C38, N17Q₄₀C38, N17Q₅₆C38 variants. The radius of gyration was normalized by $1/N^{0.5}$ where N is the number of residues. We found that the removal of the C38 domain leads to polyQ R_g^* distributions that are similar to those seen for the longer pathogenic Q-lengths (**Fig. 4b**). Specifically, the N17Q₄₀K₂ and Q₄₀K₂ R_g^* distributions coincide closely with the distribution of N17Q₅₆C38, all of which show tight distributions that sample more compact structures. By contrast, the N17Q₄₀C38 and Q₄₀C38 proteins show broader distributions, as expected. Interestingly, the N17Q₂₂C38 protein shows the broadest normalized distribution and is able to sample elongated structures that are not sampled by any of the other proteins, including the full proteins and domain deletion variants. As was done in **Fig. 4a**, we have highlighted this non-overlap region sampled by the N17Q₂₂C38 protein using a dashed box. The right side of **Fig. 4b** shows the representative structure sampled in the boxed area, which reveals that the N17 domain interacts mostly with the C38 domain and does not interact much with the polyQ. The polyQ domain also mostly interacts with the C38 domain, which likely further stabilizes the elongated structure. Again, this suggests that interactions between the C38 and polyQ domains help to stabilize elongated structures that are less likely to form β -rich motifs.

Meanwhile, since the N17 domain mostly interacts with the C38 region, it is less likely to form structures that stabilize the β -sheets in the polyQ region. Previous experimental FRET results have shown that the distance between the N17 and C38 domains increases with the Q-length⁴¹. The increased distance likely impairs the ability of C38 to disrupt contacts between the N17 and polyQ regions, consistent with our current simulation results.

The unnormalized R_g for the *entire* protein increases as expected with increasing number of residues, except for the N17Q₂₂C38 case, which samples a broad distribution of less compact structures (**Fig. S1a**). When comparing the normalized R_g^* for the full protein (not just the polyQ region) in all variants, again N17Q₂₂C38 has a broader distribution that samples less compact structures, but it is difficult to discern a clear trend from the other protein variants (**Fig. S1b**). The Q₄₀K₂ distribution does show some difference from the other non-WT proteins. The N17 and C38 R_g distributions do not show significant Q-length dependence or much of a flanking domain dependence (**Fig. S2a** and **Fig. S2b**, respectively).

N17 and polyQ cooperate to stabilize β -content

As previously shown, interactions between the C38 and N17 domains help to stabilize the extended polyQ structures. We found that the non-pathogenic “wild-type” N17Q₂₂C38 protein samples structures that have more contacts between the two domains (**Fig. S3a**) when comparing the distribution of number of contacts between N17 and C38 for the N17Q₂₂C38, N17Q₄₀C38, and N17Q₅₆C38 proteins. Contacts were defined as C $_{\alpha}$ atoms within 6 Å of each other. The sampled area of the distribution with the most number of contacts between N17 and C38 is highlighted in the dashed box. The cluster center structure from this region shows that the protein is sampling structures that have almost no interaction between the N17 and polyQ regions (**Fig. S3b**).

When the N17 domain is removed, very little difference is seen in the number of contacts between the C38 and polyQ domains (**Fig. S4**). For the full protein variants, there appears to be little Q-length dependence in the number of contacts between the polyQ and C38 regions, with the difference between N17Q₂₂C38 and the other longer Q-lengths mostly due to the total lower number of glutamine residues available to interact with the C38 domain.

When the C38 domain is removed, the N17 domain is able to sample structures with a high number of residue contacts between the N17 and polyQ domains (**Fig. 5**). To demonstrate this, the contact distributions of the N17Q₂₂C38, N17Q₄₀C38, N17Q₅₆C38, and N17Q₄₀K₂ variants were compared to each other (left panel of **Fig. 5**). As the Q-length increases for the full-protein variants we find an increase in the number of contacts between the N17 and polyQ. Another interesting finding is that all of the full protein variants have significant sampling in the region with at least ~1-2 N17-polyQ contacts, while the N17Q₄₀K₂ protein samples almost no structures in this region. Meanwhile, the most probable number of contacts for the N17Q₄₀K₂ protein, ~16 contacts, is higher than any of the distribution peaks for the full protein variants.

The dashed boxes labeled **a** and **b** for the N17Q₄₀K₂ protein in **Fig. 5** indicate the areas where this protein samples structures with significantly more N17-polyQ contacts than the other full exon-1 proteins. To the right of the distribution plot, the most representative structures from both of the highlighted areas are displayed for further illustration. The structure from area **a** illustrates that the N17 and polyQ are able to make anti β -sheet structures across both domains. In this case, the N17 and polyQ domains cooperate to stabilize the β -content on both regions. This suggests that the increase in well-formed β -sheets observed when C38 is removed is likely due to the increased number of contacts between N17 and polyQ. The representative structure from area **b** shows that the removal of the C38 can lead to structures where the polyQ envelopes the N17 domain. Both of the structures may provide additional insight into previous experimental findings, where the removal of the C38 domain leads to an increase in fibrillization and a decrease in non-fibril forming aggregates. We propose that when the N17Q₄₀K₂ structure is able to form β -sheets, it may serve as a template for fibrillization. When the N17 is enveloped in the polyQ region, it may serve to destabilize non-fibrillar aggregates by hiding the N17 domain and not allowing the polyQ to sample as many well-formed β -sheet structures.

Conclusions

In this study, we used REMD simulations to investigate how the flanking domains modulate the secondary structure motifs important for the Htt protein aggregation and fibril formation. Our results indicate that the removal of the C38 domain leads to an increase in β -sheet content, which suggests that C38 serves a protective role by destabilizing aggregate prone β -content. When the C38 domain is removed, β -sheet content on both the N17 and polyQ regions increases. The N17

acts to reinforce β -sheet content between itself and the polyQ domain, yielding well-formed β -sheets that may serve as a template for fibril formation. Although Q₄₀K₂ also shows increased β -sheet content, it does not appear to form the same type of hairpin structures that could serve as a fibril template.

Analysis of the polymeric properties of the exon-1 deletion variants compared to the full protein revealed that the C38 domain forms interactions with both the polyQ and N17 regions, leading to elongated structures in the polyQ domain. This is likely how the C38 domain is able to serve a protective, aggregate destabilizing function. When the C38 domain is present, the N17 region interacts mostly with the C38 region. At shorter Q-lengths, C38 is able to form more contacts with N17, however, as Q-length increases, the contacts between C38 and the polyQ regions seem to saturate, leading to more compact structures in the polyQ domain and fewer N17-C38 contacts. When C38 is missing, however, interactions between the N17 and polyQ regions serve to reinforce β -sheet structure

Our simulations also provide a possible molecular level explanation of the fact that the removal of the C38 domain leads to the destabilization of non-fibril forming aggregates. The structures that were found in the high-contact region of the N17-polyQ contact distribution show that β -hairpin structures are formed across the N17 and polyQ domains. These β -hairpins can likely serve as fibril forming aggregation templates. We also showed that the protein is able to form structures with many contacts between the N17 and polyQ. Where the N17 is protectively buried in the polyQ region. These buried structures lead to the formation of compact configurations without much secondary structure. We propose that the N17Q₄₀K₂ deletion variant can switch between structures that impose fibrillization through a template mechanism or disrupt β -sheet content within the protein by burying the N17 within the polyQ region. Understanding exactly how these types of structures affect the aggregation mechanism will require further work focusing on the interactions between pairs of proteins to determine how structure and oligomerization influence each other.

Methods

Molecular dynamics simulations of full length Htt exon-1 proteins (MATLEKLMKAFESLKFSQ_nP₁₁QLPQP₃QAQPLLPQPQP₁₀) featuring three different Q-lengths: N17Q₂₂C38, N17Q₄₀C38, and N17Q₅₆C38 were performed in a previous study described in an earlier paper ¹⁵. Deletion variants of the N17Q₄₀C38 Htt exon-1, were performed using the same procedure as in the previous work. Three deletion variants, with either N17 removed, C38 removed, or both removed, were simulated in this study: Q₄₀C38 (Q₄₀P₁₁QLPQP₃QAQPLLPQPQP₁₀), Q₄₀K₂, and N17Q₄₀K₂ (MATLEKLMKAFESLKFSQ₄₀K₂). The two lysine residues were added to the end of the N17Q₄₀K₂ and Q₄₀K₂ sequences to reproduce experimental setups, where the lysine improves the solubility of the protein ⁴².

Because the Htt exon-1 variants are all intrinsically disordered proteins (IDPs), temperature replica exchange molecular dynamics (REMD) ⁴³ was used to sample the folding landscape. As detailed in our previous publication ¹⁵, all of the simulations were performed with GROMACS 5.0.5 ³⁷ using the OPLS-AA/L force field ⁴⁴. A time step of 4 fs using virtual sites was employed for all simulations. Initial structures were prepared using the following procedure ¹⁵: fully extended polypeptide chains *in vacuo* were collapsed using energy minimization; the collapsed chains were equilibrated at 700 K *in vacuo*; the chains were solvated with a 2.0 nm layer of TIP3P water and neutralized with NaCl to a concentration of 100 mM; the system was heated from 298 K to 600 K and equilibrated at high temperature for 400 ns; during the final 128 ns, 128 frames were then extracted at a rate of one per ns to serve as the initial conditions for each of the REMD windows.

After equilibrating all of the windows for 1 ns, REMD using 128 replicas at constant volume was performed over a temperature range of 298 K to 600 K. The windows were all run for at least 500 ns and replicas were allowed to exchange every 1000 time steps, with an acceptance ratio in the range of 0.25-0.3.

For all analysis, the first 100 ns of each trajectory were discarded. Coordinates were recorded every 0.1 ns. Analysis was performed on the trajectories within the 298 K to 315 K temperature range. Secondary structure analyses were done using STRIDE ³⁹ and the PLUMED software package ⁴⁵. Molecular structures were visualized and rendered using VMD ⁴⁶ (<http://www.ks.uiuc.edu/Research/vmd/>).

Acknowledgements

We would like to thank Tien Huynh, Thomas F. Vogt, Amrita Mohan, Seung-gu Kang, Joseph Morrone, Alicia Preiss and Anne V. Vázquez for their various help with this work. This work was partially funded by CHDI Foundation Inc., a charitable foundation that funds research into Huntington's disease. RZ acknowledges the support of the IBM Blue Gene Science Program (W1258591, W1464125, W1464164).

Figures

Fig. 1

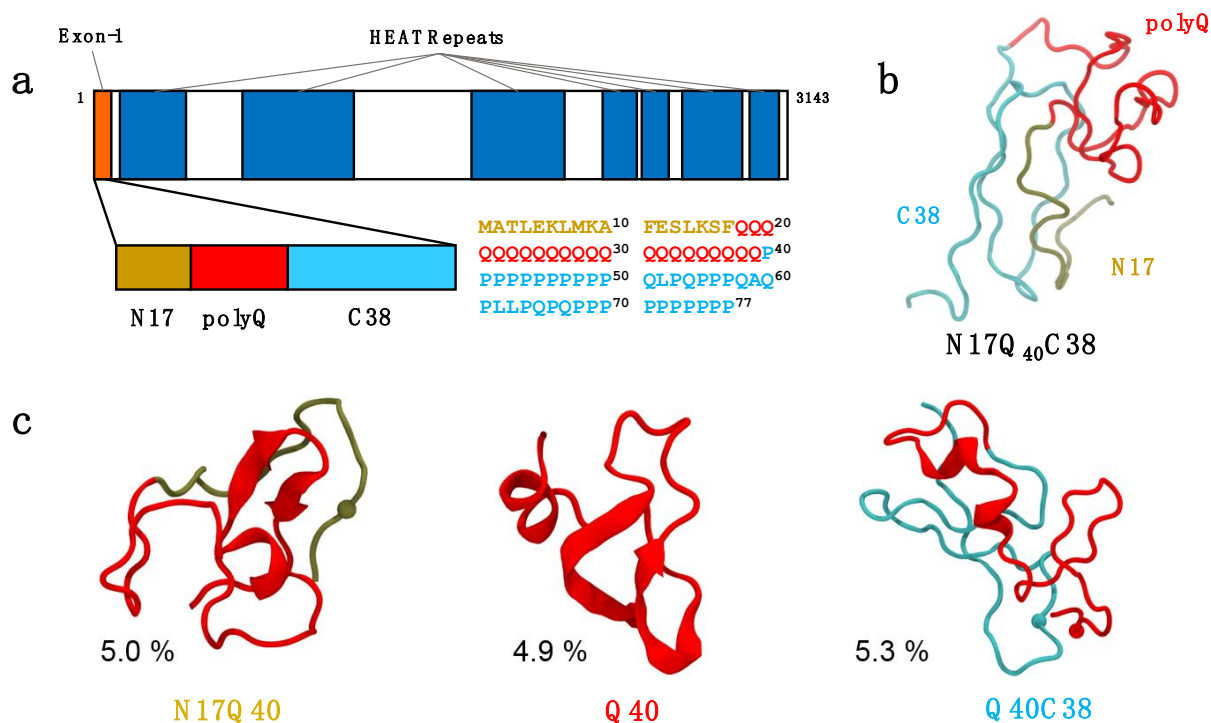


Fig. 1: The full Htt protein (a) is a very large protein consisting of many structured HEAT repeat domains along with the N-terminal Htt exon-1 domain. Huntington's disease is primarily associated with an expansion of the polyQ region of the Htt exon-1 domain. The center structure from the most occupied N17Q₄₀C38 cluster (b) further illustrates the disordered nature of the Htt exon-1 domain, even at Q-lengths past the pathogenic threshold length. Cluster center structures of the N17Q₄₀K₂, Q₄₀K₂, and Q₄₀C38 structures (c) show that while the removal of the C38 domain leads (cyan) to an increase in β -content, the N17 domain (tan) seems to reinforce the formation of well-formed β -sheets in the polyQ region (red). The structures are shown using the New Cartoon representation with N17 in tan, polyQ in red, and C38 in cyan. The spheres represent residues adopting a β -bridge structure.

Fig 2

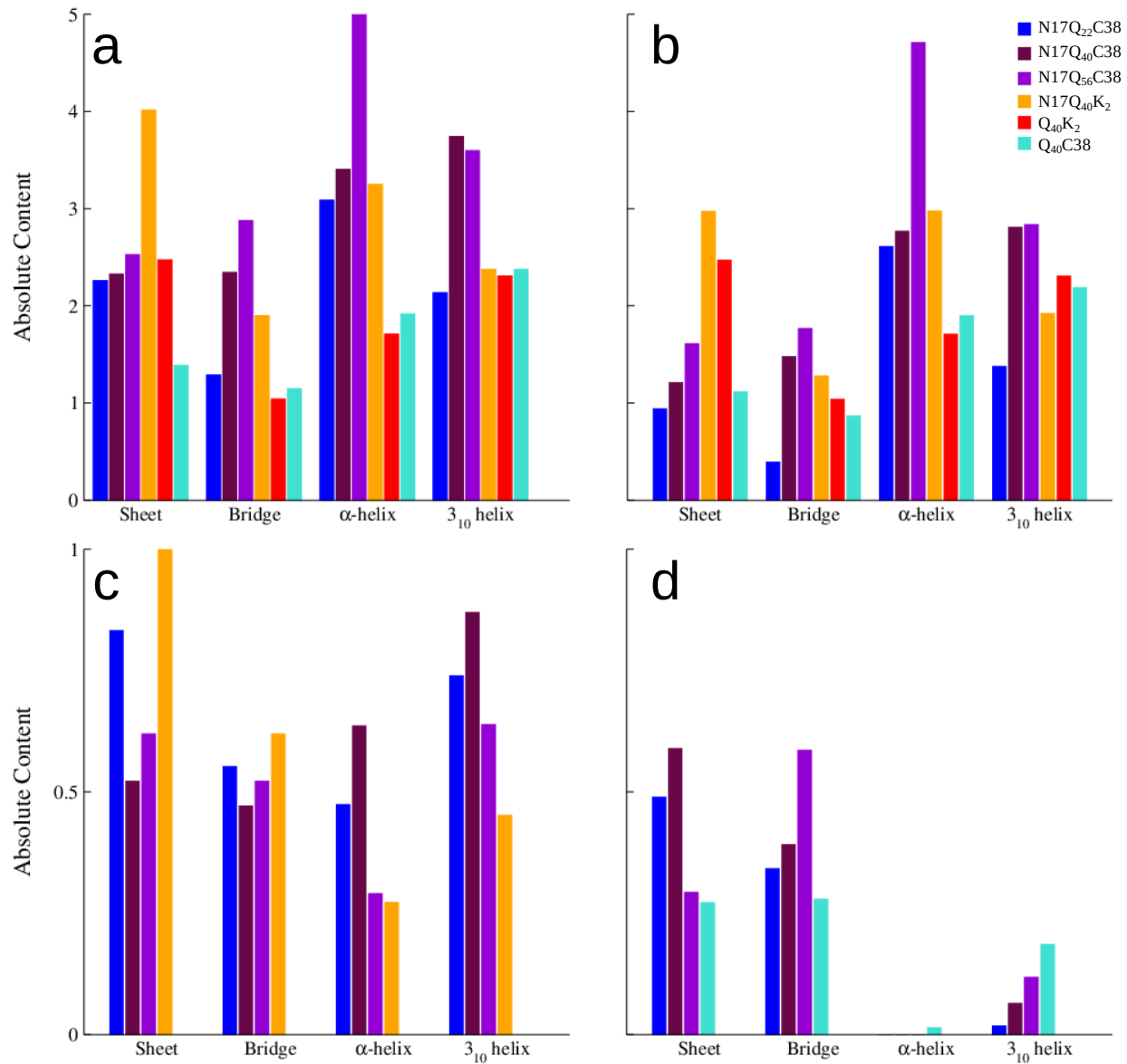


Fig. 2: Histogram plots of the secondary structure content of the Htt exon-1 variants for the full protein (a), the polyQ region (b), the N17 region (c), and the C38 region (d) calculated using the STRIDE assignment algorithm. When N17 does not interact with the C38 region (N17Q₄₀K₂), the β -sheet content of the entire protein is larger than any of the other Htt exon-1 variants, including the Q₅₆ case. This increase arises from both β -sheet content in the polyQ and N17 regions, suggesting an interaction between those two domains stabilize the β -sheet content on both regions.

Fig 3

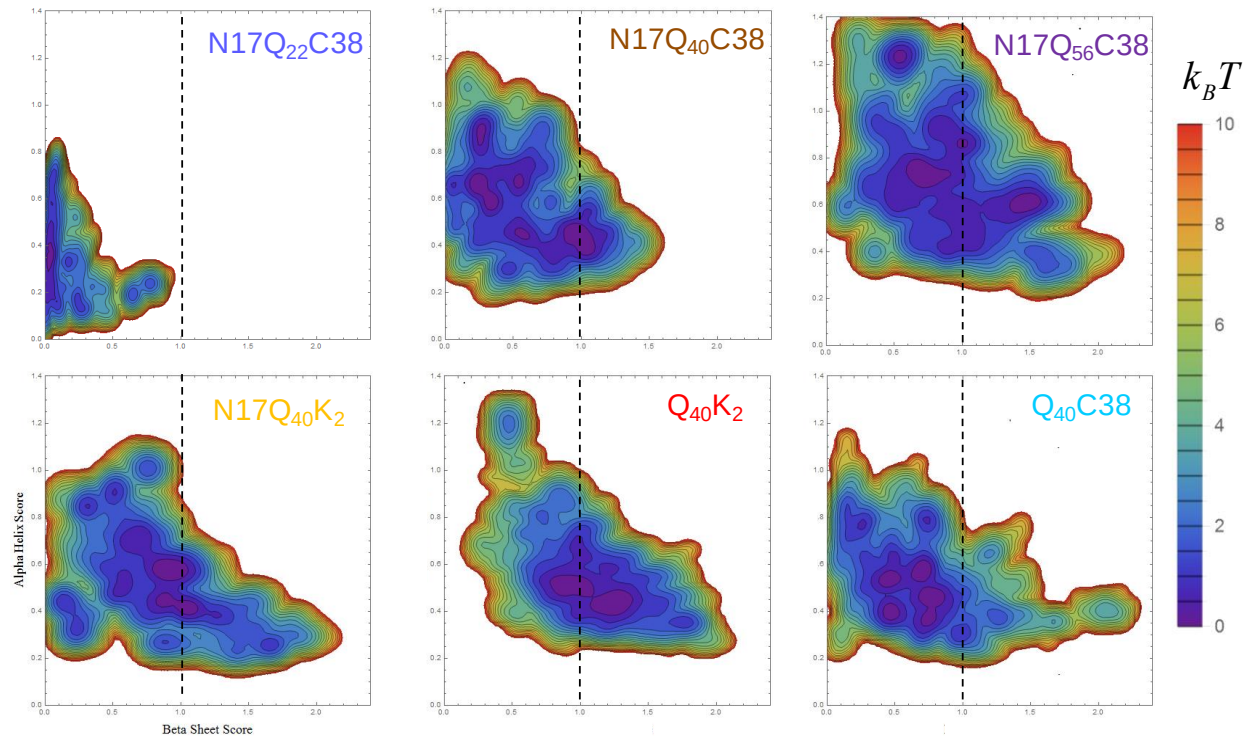


Fig. 3: The potential of mean force plots for β -sheet content and α -helix content in the polyQ domain show that when the N17 domain is removed, the polyQ region is not able to sample as much β -sheet content. The Q₄₀C38 plot shows that energy wells are shifted below a β -sheet score of one, which corresponds to an ideal six-residue sheet and is marked on the plots by a dotted vertical line. When compared to the N17Q₄₀K₂, Q₄₀K₂, and N17Q₄₀C38 cases, the Q₄₀C38 case shows a lower propensity to form β -sheets. The N17Q₄₀K₂ plot shows that the N17 region seems to help stabilize 6-residue β -motifs in the polyQ, while removal of both the C38 and N17 regions leads to the formation of large β -sheet motifs.

Fig. 4

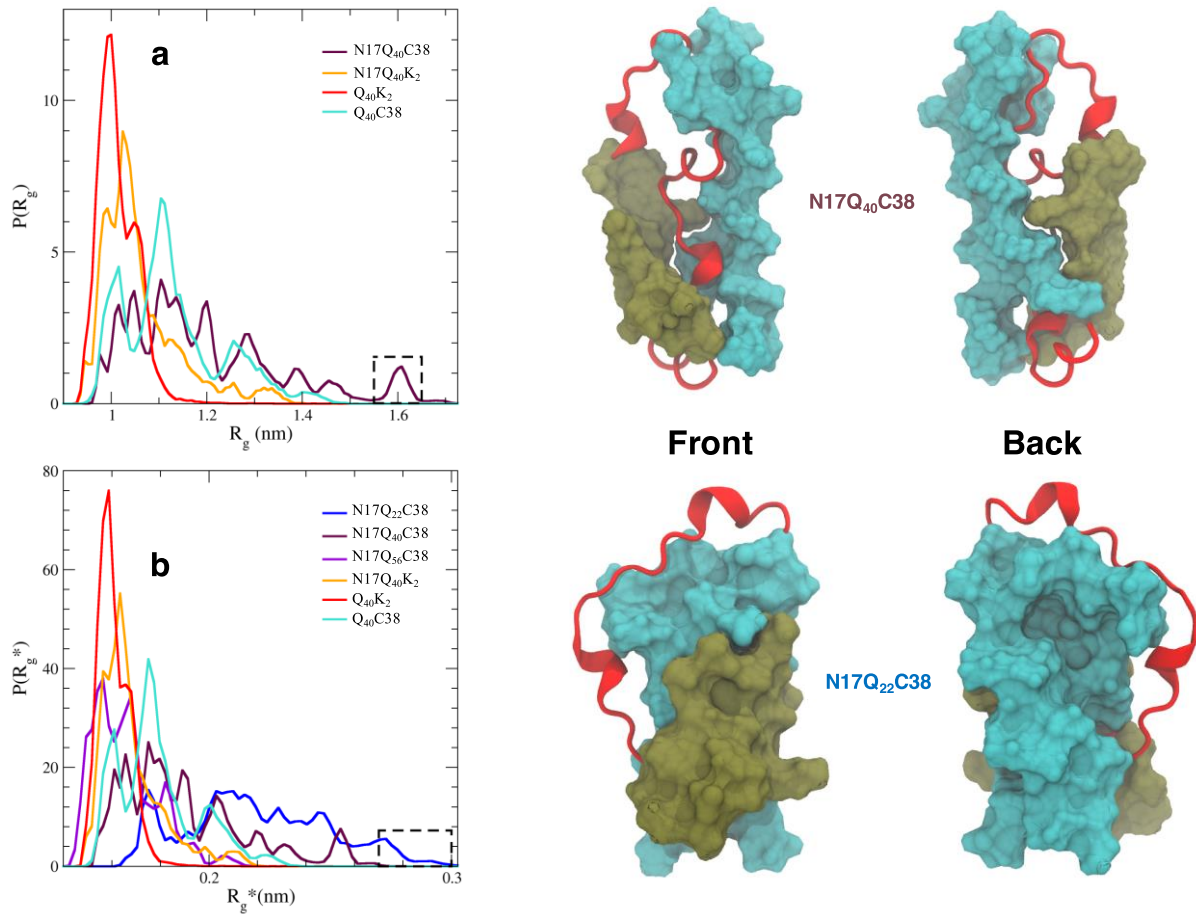


Fig. 4: Distribution plots of the R_g of the polyQ region for (a) N17Q₄₀C38, N17Q₄₀K₂, Q₄₀ K₂, and Q₄₀C38 show that interactions between the N17 and C38 domains lead to a stabilization of elongated polyQ structures that are not seen in the domain deletion variants. The shown structures, which are sampled from the boxed region, show that the polyQ is able to sample elongated helical structures when there are contacts between N17 and C38. Additionally, when comparing the normalized R_g^* of the polyQ region for multiple full length exon-1 proteins (b), the N17Q₂₂C38 variant is able to sample structures where N17 and polyQ are mostly in contact with C38 instead of each other. The highlighted structures from the boxed region that is not sampled by the longer Q-lengths shows that C38 stabilizes elongated and partially helical polyQ structures. The protein structures are shown using the New Cartoon representation for the polyQ region, shown in red, and the surf representation for the N17 and C38 regions, shown in tan and cyan, respectively.

Fig. 5

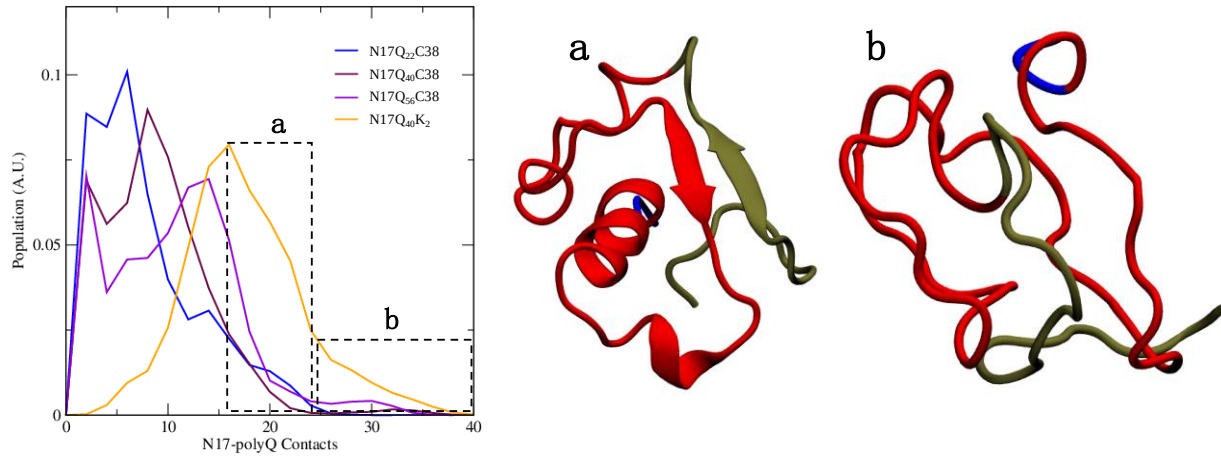


Fig. 5: The distribution of contacts between N17 and polyQ shows that when the C38 domain is removed, the N17Q₄₀K₂ protein forms structures with significantly more contacts between the N17 and polyQ domains. The cluster center structures sampled near the distribution peak (a) show that the N17 and polyQ domains are able to form β -sheets between the two regions. Additionally, the N17Q₄₀K₂ protein is also able to sample structures that maximize the number of contacts between N17 and polyQ by embedding the N17 domain within the polyQ domain (b).

References

1. Group, T. H. s. D. C. R., A novel gene containing a trinucleotide repeat that is expanded and unstable on Huntington's disease chromosomes *Cell* **1993**, 72 (6), 971 - 983.
2. Wexler, N. S., Venezuelan kindreds reveal that genetic and environmental factors modulate Huntington's disease age of onset. *Proceedings of the National Academy of Sciences* **2004**, 101, 3498-3503.
3. Gray, M.; Shirasaki, D. I.; Cepeda, C.; Andre, V. M.; Wilburn, B.; Lu, X.-H.; Tao, J.; Yamazaki, I.; Li, S.-H.; Sun, Y. E.; Li, X.-J.; Levine, M. S.; Yang, X. W., Full-Length Human Mutant Huntingtin with a Stable Polyglutamine Repeat Can Elicit Progressive and Selective Neuropathogenesis in BACHD Mice. *The Journal of Neuroscience* **2008**, 28 (24), 6182-6195.
4. Saudou, F.; Finkbeiner, S.; Devys, D.; Greenberg, M. E., Huntingtin Acts in the Nucleus to Induce Apoptosis but Death Does Not Correlate with the Formation of Intranuclear Inclusions. *Cell* **1998**, 95 (1), 55-66.
5. Thakur, A. K.; Jayaraman, M.; Mishra, R.; Thakur, M.; Chellgren, V. M.; Byeon, I.-J. L.; Anjum, D. H.; Kodali, R.; Creamer, T. P.; Conway, J. F.; Gronenborn, A. M.; Wetzel, R., Polyglutamine disruption of the huntingtin exon 1 N terminus triggers a complex aggregation mechanism. *Nature Structural & Molecular Biology* **2009**, 16 (3), 380-389.
6. Rossetti, G.; Cossio, P.; Laio, A.; Carloni, P., Conformations of the Huntingtin N-term in aqueous solution from atomistic simulations. *FEBS Letters* **2011**, 585 (19), 3086-3089.
7. Zhang, L.; Kang, H.; Vázquez, F. X.; Toledo-Sherman, L.; Luan, B.; Zhou, R., Molecular Mechanism of Stabilizing the Helical Structure of Huntingtin N17 in a Micellar Environment. *The Journal of Physical Chemistry B* **2017**, 121 (18), 4713-4721.
8. Kelley, N. W.; Huang, X.; Tam, S.; Spiess, C.; Frydman, J.; Pande, V. S., The Predicted Structure of the Headpiece of the Huntingtin Protein and Its Implications on Huntingtin Aggregation. *Journal of Molecular Biology* **2009**, 388 (5), 919-927.
9. Kim, M. W.; Chelliah, Y.; Kim, S. W.; Otwinowski, Z.; Bezprozvanny, I., Secondary Structure of Huntingtin Amino-Terminal Region. *Structure* **2009**, 17 (9), 1205-1212.
10. Kim, M., Beta conformation of polyglutamine track revealed by a crystal structure of Huntingtin N-terminal region with insertion of three histidine residues. *Prion* **2013**, 7 (3), 221-228.
11. Michalek, M.; Salnikov, E. S.; Bechinger, B., Structure and Topology of the Huntingtin 1??17 Membrane Anchor by?a?Combined Solution and Solid-State \NMR\ Approach *Biophysical Journal* **2013**, 105 (3), 699 - 710.
12. Leili, Z.; Mei, F.; Ruhong, Z.; Binqun, L., Structural perturbations on huntingtin N17 domain during its folding on 2D-nanomaterials. *Nanotechnology* **2017**, 28 (35), 354001.
13. Stapley, B. J.; Creamer, T. P., A survey of left-handed polyproline II helices. *Protein Sci* **1999**, 8 (3), 587-595.
14. Darnell, G. D.; Derryberry, J.; Kurutz, J. W.; Meredith, S. C., Mechanism of Cis-Inhibition of PolyQ Fibrillation by PolyP: \PPII\ Oligomers and the Hydrophobic Effect *Biophysical Journal* **2009**, 97 (8), 2295 - 2305.
15. Kang, H.; Vázquez, F. X.; Zhang, L.; Das, P.; Toledo-Sherman, L.; Luan, B.; Levitt, M.; Zhou, R., Emerging β -Sheet Rich Conformations in Supercompact Huntingtin Exon-1 Mutant Structures. *Journal of the American Chemical Society* **2017**, 139 (26), 8820-8827.
16. Isas, J. M.; Langen, R.; Siemer, A. B., Solid-State Nuclear Magnetic Resonance on the Static and Dynamic Domains of Huntingtin Exon-1 Fibrils. *Biochemistry* **2015**, 54 (25), 3942-9.

17. Hoop, C. L.; Lin, H. K.; Kar, K.; Hou, Z.; Poirier, M. A.; Wetzel, R.; van der Wel, P. C., Polyglutamine amyloid core boundaries and flanking domain dynamics in huntingtin fragment fibrils determined by solid-state nuclear magnetic resonance. *Biochemistry* **2014**, 53 (42), 6653-66.
18. Hoop, C. L.; Lin, H. K.; Kar, K.; Magyarfalvi, G.; Lamley, J. M.; Boatz, J. C.; Mandal, A.; Lewandowski, J. R.; Wetzel, R.; van der Wel, P. C., Huntingtin exon 1 fibrils feature an interdigitated beta-hairpin-based polyglutamine core. *Proceedings of the National Academy of Sciences of the United States of America* **2016**, 113 (6), 1546-51.
19. Lin, H. K.; Boatz, J. C.; Krabbendam, I. E.; Kodali, R.; Hou, Z.; Wetzel, R.; Dolga, A. M.; Poirier, M. A.; van der Wel, P. C. A., Fibril polymorphism affects immobilized non-amyloid flanking domains of huntingtin exon1 rather than its polyglutamine core. *Nature communications* **2017**, 8, 15462.
20. Isas, J. M.; Langen, A.; Isas, M. C.; Pandey, N. K.; Siemer, A. B., Formation and Structure of Wild Type Huntingtin Exon-1 Fibrils. *Biochemistry* **2017**, 56 (28), 3579-3586.
21. Miller, J.; Arrasate, M.; Brooks, E.; Libeu, C. P.; Legleiter, J.; Hatters, D.; Curtis, J.; Cheung, K.; Krishnan, P.; Mitra, S.; Widjaja, K.; Shaby, B. A.; Lotz, G. P.; Newhouse, Y.; Mitchell, E. J.; Osmand, A.; Gray, M.; Thulasiramin, V.; Frederic; Segal, M.; Yang, X. W.; Masliah, E.; Thompson, L. M.; Muchowski, P. J.; Weisgraber, K. H.; Finkbeiner, S., Identifying polyglutamine protein species in situ that best predict neurodegeneration. *Nature Chemical Biology* **2011**, 7, 925-934.
22. Perutz, M. F.; Johnson, T.; Suzuki, M.; Finch, J. T., Glutamine repeats as polar zippers: Their possible role in inherited neurodegenerative diseases. *Proceedings of the National Academy of Sciences* **1994**, 91, 5355-5358.
23. Crick, S. L.; Jayaraman, M.; Frieden, C.; Wetzel, R.; Pappu, R. V., Fluorescence correlation spectroscopy shows that monomeric polyglutamine molecules form collapsed structures in aqueous solutions. *Proceedings of the National Academy of Sciences* **2006**, 103 (45), 16764-16769.
24. Miettinen, M. S.; Knecht, V.; Monticelli, L.; Ignatova, Z., Assessing Polyglutamine Conformation in the Nucleating Event by Molecular Dynamics Simulations. *The Journal of Physical Chemistry B* **2012**, 116 (34), 10259-10265.
25. Heck, B. S.; Doll, F.; Hauser, K., Length-dependent conformational transitions of polyglutamine repeats as molecular origin of fibril initiation. *Biophysical Chemistry* **2014**, 185, 47-57.
26. Hoop, C. L.; Lin, H.-K.; Kar, K.; Magyarfalvi, G.; Lamley, J. M.; Boatz, J. C.; Mandal, A.; Lewandowski, J. R.; Wetzel, R.; van der Wel, P. C. A., Huntingtin exon 1 fibrils feature an interdigitated β -hairpin-based polyglutamine core. *Proceedings of the National Academy of Sciences* **2016**, 113 (6), 1546-1551.
27. Chen, S.; Ferrone, F. A.; Wetzel, R., Huntington's disease age-of-onset linked to polyglutamine aggregation nucleation. *Proceedings of the National Academy of Sciences* **2002**, 99 (18), 11884-11889.
28. Crick, S. L.; Ruff, K. M.; Garai, K.; Frieden, C.; Pappu, R. V., Unmasking the roles of N- and C-terminal flanking sequences from exon 1 of huntingtin as modulators of polyglutamine aggregation. *Proceedings of the National Academy of Sciences of the United States of America* **2013**, 110 (50), 20075-20080.

29. Duennwald, M. L.; Jagadish, S.; Muchowski, P. J.; Lindquist, S., Flanking sequences profoundly alter polyglutamine toxicity in yeast. *Proceedings of the National Academy of Sciences of the United States of America* **2006**, *103* (29), 11045-50.
30. Thakur, A. K.; Jayaraman, M.; Mishra, R.; Thakur, M.; Chellgren, V. M.; Byeon, I. J.; Anjum, D. H.; Kodali, R.; Creamer, T. P.; Conway, J. F.; Gronenborn, A. M.; Wetzel, R., Polyglutamine disruption of the huntingtin exon 1 N terminus triggers a complex aggregation mechanism. *Nat Struct Mol Biol* **2009**, *16* (4), 380-9.
31. Tam, S.; Spiess, C.; Auyeung, W.; Joachimiak, L.; Chen, B.; Poirier, M. A.; Frydman, J., The chaperonin TRiC blocks a huntingtin sequence element that promotes the conformational switch to aggregation. *Nat Struct Mol Biol* **2009**, *16* (12), 1279-85.
32. Williamson, T. E.; Vitalis, A.; Crick, S. L.; Pappu, R. V., Modulation of polyglutamine conformations and dimer formation by the N-terminus of huntingtin. *J Mol Biol* **2010**, *396* (5), 1295-309.
33. Ruff, K. M.; Khan, S. J.; Pappu, R. V., A coarse-grained model for polyglutamine aggregation modulated by amphipathic flanking sequences. *Biophys J* **2014**, *107* (5), 1226-1235.
34. Vitalis, A.; Pappu, R. V., ABSINTH: A new continuum solvation model for simulations of polypeptides in aqueous solutions. *Journal of computational chemistry* **2009**, *30* (5), 673-699.
35. Fodale, V.; Kegulian, N. C.; Verani, M.; Cariulo, C.; Azzollini, L.; Petricca, L.; Daldin, M.; Boggio, R.; Padova, A.; Kuhn, R.; Pacifici, R.; Macdonald, D.; Schoenfeld, R. C.; Park, H.; Isas, J. M.; Langen, R.; Weiss, A.; Caricasole, A., Polyglutamine- and Temperature-Dependent Conformational Rigidity in Mutant Huntingtin Revealed by Immunoassays and Circular Dichroism Spectroscopy. *PLOS ONE* **2014**, *9* (12), e112262.
36. Kar, K.; Hoop, C. L.; Drombosky, K. W.; Baker, M. A.; Kodali, R.; Arduini, I.; van der Wel, P. C. A.; Horne, W. S.; Wetzel, R., β -hairpin-mediated nucleation of polyglutamine amyloid formation. *Journal of molecular biology* **2013**, *425* (7), 1183-1197.
37. Berendsen, H. J. C.; van der Spoel, D.; van Drunen, R., GROMACS: A message-passing parallel molecular dynamics implementation *Computer Physics Communications* **1995**, *91* (1??3), 43 - 56.
38. Daura, X.; Gademann, K.; Jaun, B.; Seebach, D.; van Gunsteren, W. F.; Mark, A. E., Peptide Folding: When Simulation Meets Experiment. *Angewandte Chemie International Edition* **1999**, *38* (1-2), 236-240.
39. Frishman, D.; Argos, P., Knowledge-based protein secondary structure assignment. *Proteins: Structure, Function, and Bioinformatics* **1995**, *23* (4), 566-579.
40. Pietrucci, F.; Laio, A., A Collective Variable for the Efficient Exploration of Protein Beta-Sheet Structures: Application to SH3 and GB1. *Journal of Chemical Theory and Computation* **2009**, *5* (9), 2197-2201.
41. Caron, N. S.; Desmond, C. R.; Xia, J.; Truant, R., Polyglutamine domain flexibility mediates the proximity between flanking sequences in huntingtin. *Proceedings of the National Academy of Sciences of the United States of America* **2013**, *110* (36), 14610-5.
42. Crick, S. L.; Jayaraman, M.; Frieden, C.; Wetzel, R.; Pappu, R. V., Fluorescence correlation spectroscopy shows that monomeric polyglutamine molecules form collapsed structures in aqueous solutions. *Proceedings of the National Academy of Sciences of the United States of America* **2006**, *103* (45), 16764-9.
43. Sugita, Y.; Okamoto, Y., Replica-exchange molecular dynamics method for protein folding *Chemical Physics Letters* **1999**, *314* (1??2), 141 - 151.

44. Kaminski, G. A.; Friesner, a. R. A.; Tirado-Rives, J.; Jorgensen, W. L., Evaluation and Reparametrization of the OPLS-AA Force Field for Proteins via Comparison with Accurate Quantum Chemical Calculations on Peptides?? *The Journal of Physical Chemistry B* **2001**, *105* (28), 6474-6487.
45. Bonomi, M.; Branduardi, D.; Bussi, G.; Camilloni, C.; Provasi, D.; Raiteri, P.; Donadio, D.; Marinelli, F.; Pietrucci, F.; Broglia, R. A.; Parrinello, M., PLUMED: A portable plugin for free-energy calculations with molecular dynamics. *Computer Physics Communications* **2009**, *180* (10), 1961-1972.
46. Humphrey, W.; Dalke, A.; Schulten, K., VMD: visual molecular dynamics. *Journal of molecular graphics* **1996**, *14* (1), 33-8, 27-8.